05/08/2026

De-equivariantization

Last time. Sketched the proof that for a (smooth) projective fan (N, Σ), the functor

$$K_\Sigma : \mathrm{QCoh}(X_\Sigma / T) \longrightarrow \mathrm{Sh}_{\Lambda_\Sigma}(M_\mathbb{R})$$

is an equivalence.

Today. The non-equivariant version and the de-equivariant-ization trick.

Non-equivariant statement

Ntn. Write $q : M_{\mathbb{R}} \longrightarrow M_{\mathbb{R}}/M$ for the quotient map and

$$\bar{\Lambda}_\Sigma := dq(\Lambda_\Sigma) \simeq \bigcup_{\sigma \in \Sigma} (\sigma^\perp / (\sigma^\perp \cap M)) \times (-\sigma)$$

in

$$T^\vee(M_{\mathbb{R}}/M) \simeq (M_{\mathbb{R}}/M) \times N_{\mathbb{R}}.$$

Note that the quotient map q is both open and closed. Thus for each stratum $S \subset M_{\mathbb{R}}$ for the FLTZ stratification, $q(S)$ is locally closed. The images $q(S)$ for S a stratum of the FLTZ stratification define a stratification of $M_{\mathbb{R}}/M$ that we denote by $\bar{P}_\Sigma$.

> It is easy to see that $Sh_{\bar{\Lambda}_\Sigma}(M_{\mathbb{R}}/M) \subset Cons_{\bar{P}_\Sigma}(M_{\mathbb{R}}/M)$

Thm. Let (N, Σ) be a (smooth) projective fan. There is a natural fully faithful symmetric monoidal left adjoint

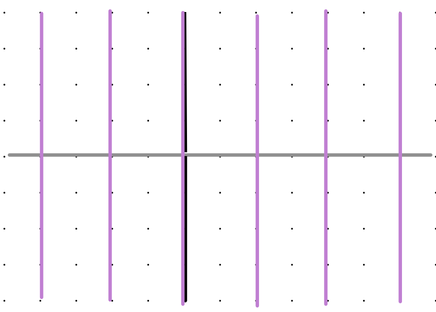
$$\bar{\kappa}_\Sigma : (\mathrm{QCoh}(X_\Sigma), \otimes) \longrightarrow (\mathrm{Sh}(M_{\mathbb{R}}/M), *).$$

Moreover, $\mathrm{im}(\bar{\kappa}_\Sigma) = \mathrm{Sh}_{\bar{\Lambda}_\Sigma}(M_{\mathbb{R}}/M)$.

Beilinson's description of $QCoh(\mathbb{P}^1)$

> Before explaining the proof, let's do an example.

Ex. Let's give the simplest examples of these results.
Let $N = \mathbb{Z}$ and let Σ be the cone generated by e_1 and e_2.
So $X_\Sigma = \mathbb{P}^1$ and $T = \mathbb{G}_m$. In this case Λ_Σ is the Lagrangian

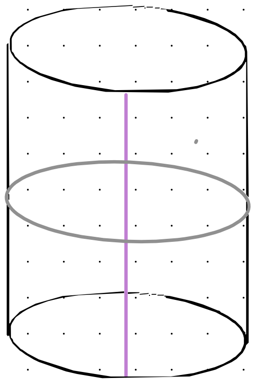


given by the union of conormals to the FLTZ stratification.

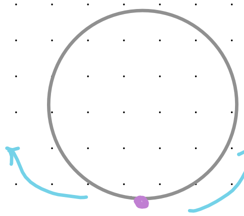
So the microlocal characterization of constructible sheaves tells us that $Sh_{\Lambda_\Sigma}(\mathbb{R}) = Cons_{P_\Sigma}(\mathbb{R})$ and exodromy tells us that

$$QCoh(\mathbb{P}^1) \simeq Fun(\cdots \leftarrow \bullet \rightarrow \bullet \leftarrow \bullet \rightarrow \cdots, Sp)$$

We have $M_{\mathbb{R}}/M = \mathbb{R}/\mathbb{Z} = S^1$. In this case, $\overline{\Lambda}_\Sigma$ is



Also the union of conormals to the stratification



In this case, a fancier version of exodromy tells us that

$$QCoh(\mathbb{P}^1) \simeq Cons_{\overline{P}_\Sigma}(S^1) \simeq Fun(\bullet \rightrightarrows \bullet, Sp)$$

Relative tensor products

Obs Let $\mathcal{A} \in \mathrm{CAlg}(\mathrm{Pr}^L)$ and let $\mathcal{B}, \mathcal{C} \in \mathrm{Mod}_{\mathcal{A}}(\mathrm{Pr}^L)$. Then by our earlier discussion on relative tensor products, we can form the relative tensor product

$$\mathcal{B} \underset{\mathcal{A}}{\otimes} \mathcal{C}$$

as the geometric realization of the bar construction

> In particular, given maps $\mathcal{A} \longrightarrow \mathcal{B}$ and $\mathcal{A} \longrightarrow \mathcal{C}$ in $\mathrm{CAlg}(\mathrm{Pr}^L)$, we can form $\mathcal{B} \otimes_{\mathcal{A}} \mathcal{C}$, which is also the pushout in $\mathrm{CAlg}(\mathrm{Pr}^L)$

Example If $A \longrightarrow B$ and $A \longrightarrow C$ are maps of $\mathbb{E}_\infty$-rings, then

$$\mathrm{Mod}_B \underset{\mathrm{Mod}_A}{\otimes} \mathrm{Mod}_C \simeq \mathrm{Mod}_{B \underset{A}{\otimes} C}$$

Desire (relative Künneth formula) Given maps of (reasonable) stacks $X \longrightarrow Z \longleftarrow Y$, we have

$$\mathrm{QCoh}(X \underset{Z}{\times} Y) \simeq \mathrm{QCoh}(X) \underset{\mathrm{QCoh}(Z)}{\otimes} \mathrm{QCoh}(Y).$$

Why we care. If G is a group object in Stacks and X is a stack with a G-action, there is a pullback square of stacks

$$\begin{tikzcd} X \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & X/G \arrow[d] \\ * \arrow[r] & */G = BG \end{tikzcd}$$

Normally, we think of describing $QCoh(X/G)$ in terms of $QCoh(X)$ and $QCoh(G)$. But the formula

$$QCoh(X) \simeq QCoh(*) \underset{QCoh(BG)}{\otimes} QCoh(X/G)$$

would describe $QCoh(X)$ in terms of equivariant information.

In our situation. For a (smooth projective) fan (N, Σ), we already have a functor

$$\kappa_\Sigma : QCoh(X_\Sigma / T) \longrightarrow Sh_{\Lambda_\Sigma}(M_{\mathbb{R}}).$$

If we know the relative Künneth formula, then applying $QCoh(*) \otimes_{QCoh(BT)} (-)$ to the left-hand side gives $QCoh(X_\Sigma)$.

> But $QCoh(*) \simeq Sp \simeq Sh(*)$.

> The Fourier transform tells us that

$$QCoh(BT) \simeq Fun(M, Sp) \simeq Sh(M).$$

as a discrete group

> So it also makes sense (a priori) to apply $QCoh(*) \otimes_{QCoh(BT)} (-)$ to the right-hand side in the guise of

$$Sh(*) \otimes_{Sh(M)} (-).$$

> One might then hope that

$$Sh(*) \otimes_{Sh(M)} Sh(M_{\mathbb{R}}) \simeq Sh(M_{\mathbb{R}} / M)$$

and also that the microsupport estimate comes along for the ride

- If this is all true, then we simply deduce the non-equivariant result by tensoring.

Remark. This technique (of deducing a non-equivariant result from an equivariant one) has long been known to geometric representation theorists. It goes under the name **de-equivariantization**.

Perfect Stacks

Lem Let A be an $\mathbb{E}_\infty$-ring and $M \in \mathrm{Mod}_A$. TFAE:

(1) M is a dualizable object of Mod_A

(2) M is a compact object of Mod_A

(3) M is in the smallest full subcategory of Mod_A that contains A and is closed under finite colimits, shifts, and retracts

Moreover, Mod_A is compactly generated by the objects satisfying these equivalent conditions

Ex If A is an ordinary ring, then $M \in \mathrm{Mod}_A \simeq D(A)$ is dualizable if and only if M can be represented by a bounded complex of finite projective ordinary A-modules.

Def. Let X be a stack. An object $F \in QCoh(X)$ is **perfect** if X is dualizable. Equivalently, for every map $f: Spec(A) \to X$ where A is a connective $\mathbb{E}_\infty$-ring, the object $f^*(F) \in Mod_A$ is dualizable.

> We write $Perf(X) \subset QCoh(X)$ for the full subcategory spanned by the perfect objects.

Def. A stack X is **quasi-geometric** if:

(1) The diagonal $X \longrightarrow X \times X$ is quasi-affine (i.e., factors as an open immersion followed by an affine morphism)

(2) There exists a connective $\mathbb{E}_\infty$-ring A and a faithfully flat map $Spec(A) \longrightarrow X$.

Idea. A quasi-geometric stack is a stack that can be obtained by an action of a groupoid object on $Spec(A)$ that is flat and quasi-affine.

Remark. Being quasi-geometric implies that X can functorially be recovered from $QCoh(X)$ together with its tensor product and t-structure. This is known as **Tannaka reconstruction**.

Def A stack X is **perfect** if X is quasi-geometric and the following **equivalent** conditions hold:

(1) The inclusion $Perf(X) \hookrightarrow QCoh(X)$ extends to an equivalence
$$Ind(Perf(X)) \xrightarrow{\sim} QCoh(X).$$

(2) $QCoh(X)$ is compactly generated and the unit $\mathcal{O}_X$ is compact.

Ex Every qcqs scheme is a perfect stack.

Thm (after Ben-Zvi–Francis–Nadler). Given a cospan of perfect stacks $X \to Z \leftarrow Y$, there is a natural equivalence
$$QCoh(X \times_Z Y) \simeq QCoh(X) \otimes_{QCoh(Z)} QCoh(Y)$$

Prop. If (N, Σ) is a fan, then the stacks X_Σ, X_Σ/T, and BT are all perfect. In particular,

$$QCoh(X_\Sigma) \simeq QCoh(*) \otimes_{QCoh(BT)} QCoh(X_\Sigma/T)$$

> With sufficient understanding of the definitions, this is easy.

Equivanantization in topology

Obs. Let M be a discrete commutative monoid. Then the obvious equivalence

$$\mathrm{Fun}(M, \mathrm{Sp}) \simeq \mathrm{Sh}(M)$$

refines to a symmetric monoidal equivalence

$$(\mathrm{Fun}(M, \mathrm{Sp}), \otimes_{\mathrm{Day}}) \simeq (\mathrm{Sh}(M), *)$$

> So if M is a group, combining this with the Fourier transform we obtain symmetric monoidal equivalences

$$(\mathrm{QCoh}(BD_M), \otimes) \simeq (\mathrm{Fun}(M, \mathrm{Sp}), \otimes_{\mathrm{Day}}) \simeq (\mathrm{Sh}(M), *)$$

Lemma. The lax symmetric monoidal functor

$$\mathrm{Sh}_! : \mathrm{LCH} \xrightarrow{\mathrm{Rleg}} \mathrm{Corr}(\mathrm{LCH}) \xrightarrow{\mathrm{Sh}} \mathrm{Pr}^L$$

$$\begin{array}{ccc} X & & \mathrm{Sh}(X) \\ f \downarrow & \longmapsto & \downarrow f_! \\ Y & & \mathrm{Sh}(Y) \end{array}$$

is symmetric monoidal.

> Now we state/recall a descent result.

Def. A map of topological spaces $f: X \longrightarrow Y$ is **étale** or a **local homeomorphism** if the following equivalent conditions hold:

(1) f is open and the diagonal $\Delta_f: X \longrightarrow X \times_Y X$ is open.

(2) f is open and for each $x \in X$, there is an open neighborhood $U \subset X$ of x such that $f|_U: U \longrightarrow f(U)$ is a homeomorphism.

Ex. Open immersions and covering spaces are étale. In particular, for a lattice M, the quotient map $M_{\mathbb{R}} \longrightarrow M_{\mathbb{R}}/M$ is étale.

Lem. If $f: X \longrightarrow Y$ is étale, then $f^*: Sh(Y, An) \longrightarrow Sh(X, An)$ admits a left adjoint, hence the same is true with coefficients in any presentable ∞-category. Moreover, $f^*: Sh(Y; Sp) \longrightarrow Sh(X; Sp)$ is right adjoint to $f_!$.

Lem (étale descent). If $f: X \to Y$ is an étale surjection of spaces, then the natural diagram obtained using $*$-pullback functoriality

$$\mathrm{Sh}(Y) \xrightarrow{f^*} \mathrm{Sh}(X) \rightrightarrows \mathrm{Sh}(X \times_Y X) \substack{\rightarrow\\\rightarrow\\\rightarrow} \cdots$$

is a limit diagram in Cat_∞ (hence also in Pr^R). As a consequence, the diagram using $!$-pushforward functoriality

$$\cdots \substack{\rightarrow\\\rightarrow\\\rightarrow} \mathrm{Sh}(X \times_Y X) \rightrightarrows \mathrm{Sh}(X) \xrightarrow{f_!} \mathrm{Sh}(Y)$$

is a colimit diagram in Pr^L

Prop Let M be a lattice, let $i: M \hookrightarrow M_{\mathbb{R}}$ be the inclusion, let $p: M \to *$ be the unique map, let $q: M_{\mathbb{R}} \to M_{\mathbb{R}}/M$ be the quotient map, and let $u: * \to M_{\mathbb{R}}/M$ be the unit. Then the square

$$\begin{tikzcd} \mathrm{Sh}(M) \arrow[r, "i_!"] \arrow[d, "p_!"'] & \mathrm{Sh}(M_{\mathbb{R}}) \arrow[d, "q_!"] \\ \mathrm{Sh}(*) \arrow[r, "u_!"'] & \mathrm{Sh}(M_{\mathbb{R}}/M) \end{tikzcd}$$

is a pushout square in $\mathrm{CAlg}(\mathrm{Pr}^L)$, where each ∞-category is given the convolution symmetric monoidal structure.

Proof Sketch

Observe that the terms of the Čech nerve of the quotient map $M_{\mathbb{R}} \longrightarrow M_{\mathbb{R}}/M$ are given by

$$M_{\mathbb{R}}^{\times_{M_{\mathbb{R}}/M}[n]} \simeq M^{\times n} \times M_{\mathbb{R}}$$

Moreover, this simplicial object coincides with the simplicial object defining the action of M on $M_{\mathbb{R}}$ by translation. By construction, $Sh(*) \otimes_{Sh(M)} Sh(M_{\mathbb{R}})$ can be computed as the colimit in Pr^L of the bar-construction

$$\begin{tikzcd} \cdots \arrow[r, shift left=3] \arrow[r, shift left=1] \arrow[r, shift right=1] \arrow[r, shift right=3] & Sh(*) \otimes Sh(M)^{\otimes 2} \otimes Sh(M_{\mathbb{R}}) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & Sh(*) \otimes Sh(M) \otimes Sh(M_{\mathbb{R}}) \arrow[d, shift left] \arrow[d, shift right] \\ & & Sh(*) \otimes Sh(M_{\mathbb{R}}) \end{tikzcd}$$

By the Künneth formula, this diagram has the form

$$\cdots \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \mathrm{Sh}(M^{\times 2}\times M_{\mathbb{R}}) \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \mathrm{Sh}(M\times M_{\mathbb{R}}) \rightrightarrows \mathrm{Sh}(M_{\mathbb{R}})$$

Unwinding definitions shows that this diagram is obtained by applying $\mathrm{Sh}_!$ to the simplicial diagram $M^{\times \bullet}\times M_{\mathbb{R}}$ defining the translation action of M on $M_{\mathbb{R}}$. Since the quotient map $q\colon M_{\mathbb{R}} \longrightarrow M_{\mathbb{R}}/M$ is an étale surjection, by the above computation of the Čech nerve of q and étale descent, we see that the colimit is $\mathrm{Sh}(M_{\mathbb{R}}/M)$ $\square$

Interaction with microsupports

Observation. Let (N,Σ) be a fan and $\sigma \in \Sigma$ a cone of dim $\dim(N_{\mathbb{R}})$. Then $\sigma^{\perp} = 0$. So if Σ is proper, then since the union of the top dimensional cones is $N_{\mathbb{R}}$, we have

$$\bigcup_{m \in M, \sigma \in \Sigma_{top}} (m + \sigma^{\perp}) \times (-\sigma) = \bigcup_{m \in M} \{m\} \times N_{\mathbb{R}}$$

In particular, Λ_{Σ} contains this union. So for any sheaf $F \in Sh(M)$, the pushforward $i_*(F)$ has microsupport in Λ_{Σ}.

Lem. Let (N,Σ) be a fan. Then

(1) $q^* : Sh(M_{\mathbb{R}}/M) \longrightarrow Sh(M_{\mathbb{R}})$ sends $Sh_{\bar{\Lambda}_{\Sigma}}(M_{\mathbb{R}}/M)$ to $Sh_{\Lambda_{\Sigma}}(M_{\mathbb{R}})$

(2) $q_! : Sh(M_{\mathbb{R}}) \longrightarrow Sh(M_{\mathbb{R}}/M)$ sends $Sh_{\Lambda_{\Sigma}}(M_{\mathbb{R}})$ to $Sh_{\bar{\Lambda}_{\Sigma}}(M_{\mathbb{R}}/M)$.

(3) $F \in Sh(M_{\mathbb{R}}/M)$ has microsupport in $\bar{\Lambda}_{\Sigma}$ if and only if $q^*(F)$ has microsupport in Λ_{Σ}.

> The key here is that $q: M_{\mathbb{R}}/M \longrightarrow M$ is étale (in fact, a covering space)

- The claims can thus be checked using the local nature of the definition of the microsupport.

Cor. Let (N, Σ) be a (smooth) projective fan. Then the earlier equivalence

$$Sh(*) \otimes_{Sh(M)} Sh(M_{\mathbb{R}}) \xrightarrow{\sim} Sh(M_{\mathbb{R}}/M)$$

induces an equivalence

$$Sh(*) \otimes_{Sh(M)} Sh_{\Lambda_\Sigma}(M_{\mathbb{R}}) \xrightarrow{\sim} Sh_{\bar{\Lambda}_\Sigma}(M_{\mathbb{R}}/M)$$

Compatibility

Prop. Let (N, Σ) be a fan. Write $\pi : X_\Sigma / T \longrightarrow BT$ and $c : * \longrightarrow BT$ for the natural maps, $i : M \hookrightarrow M_{\mathbb{R}}$ for the inclusion, and $p : M \to *$ for the unique map. Then the diagram

$$\begin{tikzcd}
QCoh(X_\Sigma/T) \arrow[r, "\kappa_\Sigma"] & Sh(M_{\mathbb{R}}) \\
QCoh(BT) \arrow[u, "\pi^*"] \arrow[r, "\sim", "FT"'] \arrow[d, "c^*"'] & Sh(M) \arrow[u, "i_! = i_*"'] \arrow[d, "p_!"] \\
QCoh(*) \arrow[r, equal] & Sh(*)
\end{tikzcd}$$

canonically commutes. Moreover, if Σ is proper, then i_* factors through $Sh_{\Lambda_\Sigma}(M_{\mathbb{R}})$.

Proof

First recall that we already explained the 'moreover' part

For the bottom square, we identify $Sh(M) \simeq Fun(M, Sp)$, in which case $p_!$ is identified with the colimit functor. Also recall that we constructed the Fourier transform as an equivalence

$$\Phi_M : Fun(M, Sp) \xrightarrow{\sim} QCoh(BT),$$

so the bottom square uses the inverse equivalence. Thus it suffices to show that the triangle of symmetric monoidal left adjoints

$$\begin{tikzcd} Fun(M, Sp) \arrow[rr, "\sim", "\Phi_M"'] \arrow[dr, "colim"'] & & QCoh(BT) \arrow[dl, "c^*"] \\ & Sp & \end{tikzcd}$$

canonically commutes. Equivalently, the induced square of right adjoints

$$\begin{tikzcd} & \mathrm{Sp} \arrow[dl, "c_*"'] \arrow[dr, "\mathrm{const}"] & \\ \mathrm{QCoh}(BT) \arrow[rr, "\sim", "\Phi_M^{-1}"'] & & \mathrm{Fun}(M, \mathrm{Sp}) \end{tikzcd}.$$

But all of these functors preserve colimits, so the diagram canonically commutes by the universal property of spectra.

For the top square, we again recall that κ_Σ is the composite of a few functors. For $\sigma \in \Sigma$, write $p_\sigma : M \longrightarrow \Theta(\sigma)$ for the symmetric monoidal functor $m \mapsto m + \sigma^\vee$ and $\pi_\sigma : X_\sigma / T \longrightarrow BT$ for the natural map. We explained earlier in the course that the square

$$\begin{tikzcd} \lim\limits_{\sigma \in \Sigma^{op}} \mathrm{Fun}(\Theta(\sigma)^{op}, \mathrm{Sp}) \arrow[rr, "\Phi_\Sigma = \lim_{\sigma \in \Sigma^{op}} \Phi_\sigma", "\sim"'] & & \mathrm{QCoh}(X_\Sigma / T) \\ \mathrm{Fun}(M, \mathrm{Sp}) \arrow[u, "\lim_{\sigma \in \Sigma^{op}} p_{\sigma,\#}"] \arrow[rr, "\Phi_M", "\sim"'] & & \mathrm{QCoh}(BT) \arrow[u] \end{tikzcd}$$

canonically commutes. Since $\kappa_\Sigma = R_\Sigma \circ \Psi_\Sigma \circ \Phi_\Sigma^{-1}$ it suffices to show that

$$(*) \quad \mathrm{Fun}(M, \mathrm{Sp}) \xrightarrow{\lim_{\sigma \in \Sigma^{op}} p_{\sigma,\#}} \lim_{\sigma \in \Sigma^{op}} \mathrm{Fun}(\Theta(\sigma)^{op}, \mathrm{Sp}) \xrightarrow{\Psi_\Sigma} \mathrm{Sh}(M_{\mathbb{R}})$$

coincides with i_*. Since both functors preserve colimits, it suffices to show that their restrictions along the spectral Yoneda embeddings agree

> By definition, $M \hookrightarrow \mathrm{Fun}(M, \mathrm{Sp}) \simeq \mathrm{Sh}(M) \xrightarrow{i_*} \mathrm{Sh}(M_{\mathbb{R}})$ sends $m \in M$ to the skyscraper sheaf $m_* \mathbb{S}$.

> Unwinding definitions shows that $(*)$ sends $m \in M$ to

$$\lim_{\sigma \in \Sigma^{op}} \omega_{m+\sigma^\vee}$$

Recall that we showed that $\lim_{\sigma \in \Sigma^{op}} \omega_{\sigma^\vee} \simeq 0_*(\mathbb{S})$, which is the unit for convolution. Translating by m shows that

$$\lim_{\sigma \in \Sigma^{op}} \omega_{m+\sigma^\vee} \simeq m_*(\mathbb{S}). \qquad \square$$

Cor. If (N, Σ) is a (smooth) projective fan, then there is a natural equivalence $\bar{\kappa}_\Sigma$ fitting into a commutative square of symmetric monoidal left adjoints

$$\begin{tikzcd}
\mathrm{QCoh}(X_\Sigma/T) \arrow[r, "\kappa_\Sigma", "\sim"'] \arrow[d, "q^*"'] & \mathrm{Sh}_{\Lambda_\Sigma}(M_{\mathbb{R}}) \arrow[d, "q_!"] \\
\mathrm{QCoh}(X_\Sigma) \arrow[r, "\sim", "\bar{\kappa}_\Sigma"'] & \mathrm{Sh}_{\bar{\Lambda}_\Sigma}(M_{\mathbb{R}}/M).
\end{tikzcd}$$

Proof.

Take pushouts vertically in the diagram in $\mathrm{CAlg}(\mathrm{Pr}^L)$

$$\begin{tikzcd}
\mathrm{QCoh}(X_\Sigma/T) \arrow[r, "\kappa_\Sigma"] & \mathrm{Sh}_{\Lambda_\Sigma}(M_{\mathbb{R}}) \\
\mathrm{QCoh}(BT) \arrow[u, "\pi^*"] \arrow[r, "\sim", "FT"'] \arrow[d, "c^*"'] & \mathrm{Sh}(M) \arrow[u, "\iota_! = \iota_*"'] \arrow[d, "p_!"] \\
\mathrm{QCoh}(*) \arrow[r, equal] & \mathrm{Sh}(*)
\end{tikzcd}$$

$\square$

Addendum: Changing Coefficients

- Throughout the course, we've emphasized that working over $\mathbb{S}$ will allow us to deduce toric mirror symmetry with any coefficients
  - Let us now explain this

Remark. $(\mathrm{Sp}, \otimes) \in \mathrm{CAlg}(\mathrm{Pr}^L)$ is an idempotent algebra. A presentable ∞-category $\mathcal{C}$ is a module over Sp if and only if $\mathcal{C}$ is stable

Observation. If A is a connective E_∞-ring, then by the Kunneth formula, we have

$Y_A := Y \times_{\mathrm{Spec}(\mathbb{S})} \mathrm{Spec}(A)$

$$\mathrm{QCoh}(X_\Sigma) \otimes \mathrm{Mod}_A \simeq \mathrm{QCoh}(X_{\Sigma,A})$$

$$\mathrm{QCoh}(BT) \otimes \mathrm{Mod}_A \simeq \mathrm{QCoh}(B(T_A))$$

$$\mathrm{QCoh}(X_\Sigma / T) \otimes \mathrm{Mod}_A \simeq \mathrm{QCoh}(X_{\Sigma,A} / T_A)$$

> We also have that for a topological space Y, since Sp is an idempotent algebra,

$$\begin{aligned} Sh(Y, Sp) \otimes Mod_A &\simeq Sh(Y; An) \otimes Sp \otimes Mod_A \\ &\simeq Sh(Y; An) \otimes Mod_A \\ &\simeq Sh(Y; Mod_A) \end{aligned}$$

> Since the forgetful functor $Mod_A \longrightarrow Sp$ is conservative and preserves limits and colimits, it is immiate from the definition of microsupport that for a manifold Y and closed, $\mathbb{R}_{>0}$-conic subset $\Lambda \subset T^\vee Y$, the above equivalence restricts to an equivalence

$$Sh_\Lambda(Y; Sp) \otimes Mod_A \simeq Sh_\Lambda(Y; Mod_A).$$

Upshot For a (smooth) projective fan (N, Σ) and connective $\mathbb{E}_\infty$-ring A, tensoring the natural symmetric monoidal equivs

$$\kappa_\Sigma : (\mathrm{QCoh}(X_\Sigma / T), \otimes) \longrightarrow (\mathrm{Sh}_{\Lambda_\Sigma}(M_{\mathbb{R}}), *)$$

and

$$\bar{\kappa}_\Sigma : (\mathrm{QCoh}(X_\Sigma), \otimes) \longrightarrow (\mathrm{Sh}_{\bar{\Lambda}_\Sigma}(M_{\mathbb{R}}/M), *)$$

with Mod_A provides natural symmetric monoidal equivalences

$$\kappa_{\Sigma,A} : (\mathrm{QCoh}(X_{\Sigma,A} / T_A), \otimes) \longrightarrow (\mathrm{Sh}_{\Lambda_\Sigma}(M_{\mathbb{R}}; \mathrm{Mod}_A), *)$$

and

$$\bar{\kappa}_{\Sigma,A} : (\mathrm{QCoh}(X_{\Sigma,A}), \otimes) \longrightarrow (\mathrm{Sh}_{\bar{\Lambda}_\Sigma}(M_{\mathbb{R}}/M; \mathrm{Mod}_A), *)$$